

code

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Class: 4 MCA B

0.0.1 Question 1

Import libraries

```
[52]: import pandas as pd
import numpy as np
from datetime import datetime
import seaborn as sns
import matplotlib.pyplot as plt
```

Load the given road accident dataset

```
[2]: df = pd.read_csv("QP_Data_road_accident_dataset_with_missing.csv")
```

Display the first 10 rows

```
[4]: print("Dataset first 10 rows:")
df.head(10)
```

Dataset first 10 rows:

```
[4]:
```

	Country	Year	Month	Day of Week	Time of Day	Urban/Rural	Road Type	\
0	USA	2002	October	Tuesday	Evening	Rural	Street	
1	UK	2014	December	Saturday	Evening	Urban	Street	
2	USA	2012	July	Sunday	Afternoon	Urban	Highway	
3	UK	2017	May	Saturday	Evening	Urban	Main Road	
4	Canada	2002	July	Tuesday	Afternoon	Rural	Highway	
5	India	2010	May	Monday	Evening	Urban	Street	
6	China	2010	March	Monday	Afternoon	Rural	Street	
7	USA	2016	July	Friday	Afternoon	Rural	Main Road	
8	Japan	2014	August	Thursday	Afternoon	Rural	Highway	
9	USA	2007	April	Monday	Evening	Urban	Highway	

	Weather Conditions	Visibility Level	Number of Vehicles Involved	...	\
0	Windy	NaN	1	...	
1	Windy	168.311358	3	...	
2	Snowy	341.286506	4	...	
3	Clear	489.384536	2	...	
4	Rainy	348.344850	1	...	
5	Snowy	479.216834	2	...	
6	Foggy	386.176217	3	...	
7	Foggy	75.608688	3	...	
8	Rainy	387.828675	3	...	
9	Foggy	443.965408	3	...	

	Number of Fatalities	Emergency Response Time	Traffic Volume	\
0	2	58.625720	7412.752760	
1	1	58.041380	4458.628820	
2	4	NaN	9856.915064	
3	3	48.554014	4958.646267	
4	4	18.318250	3843.191463	
5	4	8.205994	360.951795	
6	3	17.851663	7607.804705	
7	2	46.740367	6061.407002	
8	2	NaN	3793.850542	
9	3	NaN	1140.429308	

	Road Condition	Accident Cause	Insurance Claims	Medical Cost	\
0	Wet	Weather	4	40499.856980	
1	Snow-covered	Mechanical Failure	3	6486.600073	
2	Wet	Speeding	4	29164.412980	
3	Icy	Distracted Driving	3	25797.212570	
4	Icy	Distracted Driving	8	15605.293920	
5	Dry	Speeding	7	NaN	
6	Wet	Weather	9	NaN	
7	Dry	Speeding	8	4262.755621	
8	Snow-covered	Mechanical Failure	5	37624.775980	
9	Snow-covered	Distracted Driving	0	15801.190080	

	Economic Loss	Region	Population Density
0	22072.878500	Europe	3866.273014
1	9534.399441	North America	2333.916224
2	58009.145120	South America	4408.889129
3	20907.151300	Australia	2810.822423
4	13584.060760	South America	3883.645634
5	45995.605250	South America	3626.074027
6	52342.431810	Asia	3408.182341
7	70652.223520	South America	408.296453
8	13724.630950	Europe	2058.898279
9	61948.862750	Australia	1840.206143

[10 rows x 30 columns]

List all numerical and categorical variables separately

```
[21]: numerical_vars = df.select_dtypes(include=['int64', 'float64']).columns.tolist()
      categorical_vars = df.select_dtypes(include=['str']).columns.tolist()

      print("Numerical variables:", numerical_vars)
      print("Categorical variables:", categorical_vars)
```

Numerical variables: ['Year', 'Visibility Level', 'Number of Vehicles Involved', 'Speed Limit', 'Driver Alcohol Level', 'Driver Fatigue', 'Pedestrians Involved', 'Cyclists Involved', 'Number of Injuries', 'Number of Fatalities', 'Emergency Response Time', 'Traffic Volume', 'Insurance Claims', 'Medical Cost', 'Economic Loss', 'Population Density']

Categorical variables: ['Country', 'Month', 'Day of Week', 'Time of Day', 'Urban/Rural', 'Road Type', 'Weather Conditions', 'Driver Age Group', 'Driver Gender', 'Vehicle Condition', 'Accident Severity', 'Road Condition', 'Accident Cause', 'Region']

Check the data types of all columns and interpret whether any column needs type conversion

```
[27]: print("Data types:")
      print(df.dtypes)
```

Data types:

Country	str
Year	int64
Month	str
Day of Week	str
Time of Day	str
Urban/Rural	str
Road Type	str
Weather Conditions	str
Visibility Level	float64
Number of Vehicles Involved	int64
Speed Limit	int64
Driver Age Group	str
Driver Gender	str
Driver Alcohol Level	float64
Driver Fatigue	int64
Vehicle Condition	str
Pedestrians Involved	int64
Cyclists Involved	int64
Accident Severity	str
Number of Injuries	int64
Number of Fatalities	int64

```

Emergency Response Time    float64
Traffic Volume             float64
Road Condition             str
Accident Cause            str
Insurance Claims           int64
Medical Cost              float64
Economic Loss             float64
Region                   str
Population Density        float64
dtype: object

```

```

[54]: # Type conversion check
      # 'Year' is numeric but can be treated as categorical (no mathematical meaning)
      # 'Insurance Claims' appears numeric - verify

      print("\nUnique values in 'Insurance Claims':", df['Insurance Claims'].
            ↪unique()[:10])

```

Unique values in 'Insurance Claims': [4 3 8 7 9 5 0 2 1 6]

0.0.2 Question 2

- Generate descriptive stats for all numerical variables. Which variable shows the highest variation

```

[58]: # Descriptive statistics for numerical variables
      desc_stats = df[numerical_vars].describe()
      print(desc_stats)

      # Variable with highest variation (standard deviation)
      highest_var = desc_stats.loc['std'].idxmax()
      print(f"\nVariable with highest variation: {highest_var}")

```

	Emergency Response Time	Medical Cost	Economic Loss \
count	118824.000000	106979.000000	118800.000000
mean	32.486666	22999.326535	46028.388017
std	15.900793	13309.840366	26642.146034
min	5.000177	500.110091	1000.335085
25%	18.694200	11594.699610	23253.528405
50%	32.549514	22687.634120	45484.381415
75%	46.288919	33869.677745	67731.981662
max	59.999588	49999.930130	99999.622970

	Driver Alcohol Level
count	104893.000000
mean	0.110463

```
std          0.066818
min          0.000002
25%          0.053893
50%          0.107667
75%          0.161436
max          0.249989
```

Variable with highest variation: Economic Loss

- b. Compare the mean and median of Medical Cost. What does the difference indicate about Skewness?

```
[29]: medical_cost_mean = df['Medical Cost'].mean()
      medical_cost_median = df['Medical Cost'].median()

      print(f"Mean of Medical Cost: {medical_cost_mean}")
      print(f"Median of Medical Cost: {medical_cost_median}")

      # Interpretation
      if medical_cost_mean > medical_cost_median:
          print("Skewness: Right-skewed (positive skew)")
      else:
          print("Skewness: Left-skewed (negative skew)")
```

Mean of Medical Cost: 22999.326534801297

Median of Medical Cost: 22687.63412

Skewness: Right-skewed (positive skew)

0.0.3 Question 3

- a. Identify all variables containing missing values. Calculate the number and percentage of missing values in each column

```
[30]: # Missing values count and percentage
      missing_values = df.isnull().sum()
      missing_percentage = (missing_values / len(df)) * 100

      missing_df = pd.DataFrame({
          'Missing Count': missing_values,
          'Missing Percentage': missing_percentage
      }).sort_values(by='Missing Count', ascending=False)

      print(missing_df[missing_df['Missing Count'] > 0])
```

	Missing Count	Missing Percentage
Driver Alcohol Level	27107	20.535606
Medical Cost	25021	18.955303
Economic Loss	13200	10.000000

Emergency Response Time	13176	9.981818
Population Density	6600	5.000000
Visibility Level	6600	5.000000
Traffic Volume	6600	5.000000

- b. Explain and identify which missing value technique can be used for variables Visibility Level, Traffic Volume and Population Density

Ans.

For Visibility Level, Traffic Volume, Population Density: - Use mean/median imputation if missingness is random (MCAR/MAR). - Example: `df['Visibility Level'].fillna(df['Visibility Level'].mean(), inplace=True)`

- c. Explain and identify which missing value technique can be used for variables Driver Alcohol Level, Emergency Response Time and Medical Cost

Ans.

For Driver Alcohol Level, Emergency Response Time, Medical Cost: - Driver Alcohol Level: If missing due to non-reporting (MNAR), consider mode imputation or flag missingness. - Emergency Response Time / Medical Cost: Use median imputation (robust to outliers). - Example: `df['Medical Cost'].fillna(df['Medical Cost'].median(), inplace=True)`

- d. How can missingness in Driver Alcohol Level be both MAR and MNAR? Explain with reference to the dataset

Ans.

MAR (Missing At Random): Missingness may depend on observed data (e.g., accidents with no alcohol test due to minor severity).

MNAR (Missing Not At Random): Missingness may depend on unobserved data (e.g., drivers hide alcohol consumption).

In this dataset, missingness could be both: - MAR: If missingness correlates with other variables (e.g., Accident Severity). - MNAR: If drivers with high alcohol levels are less likely to report.

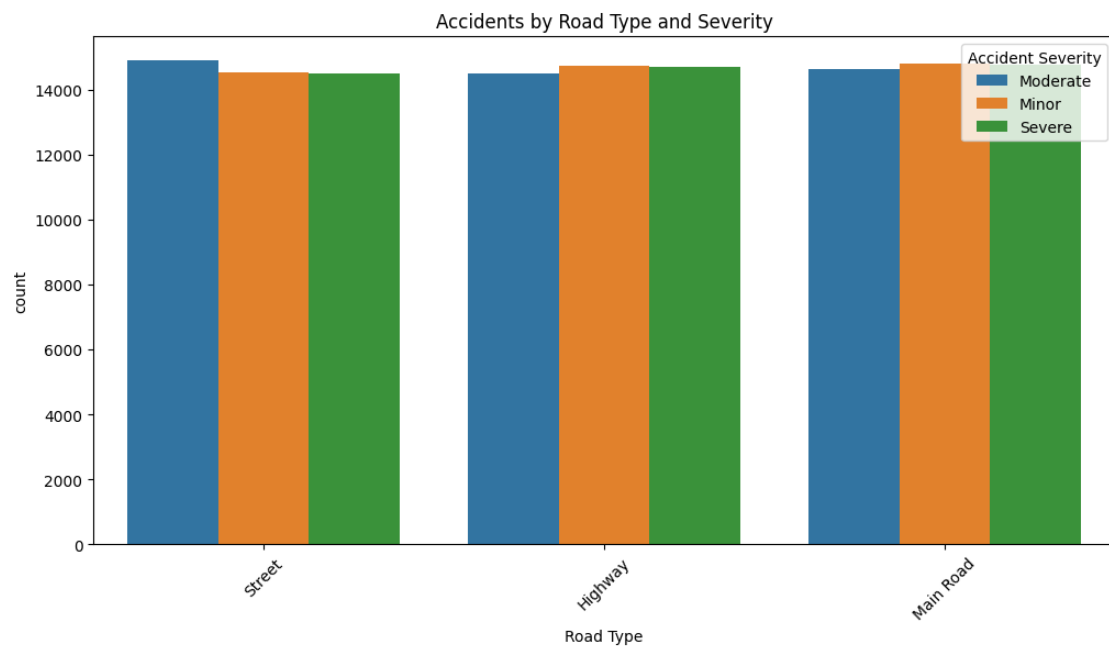
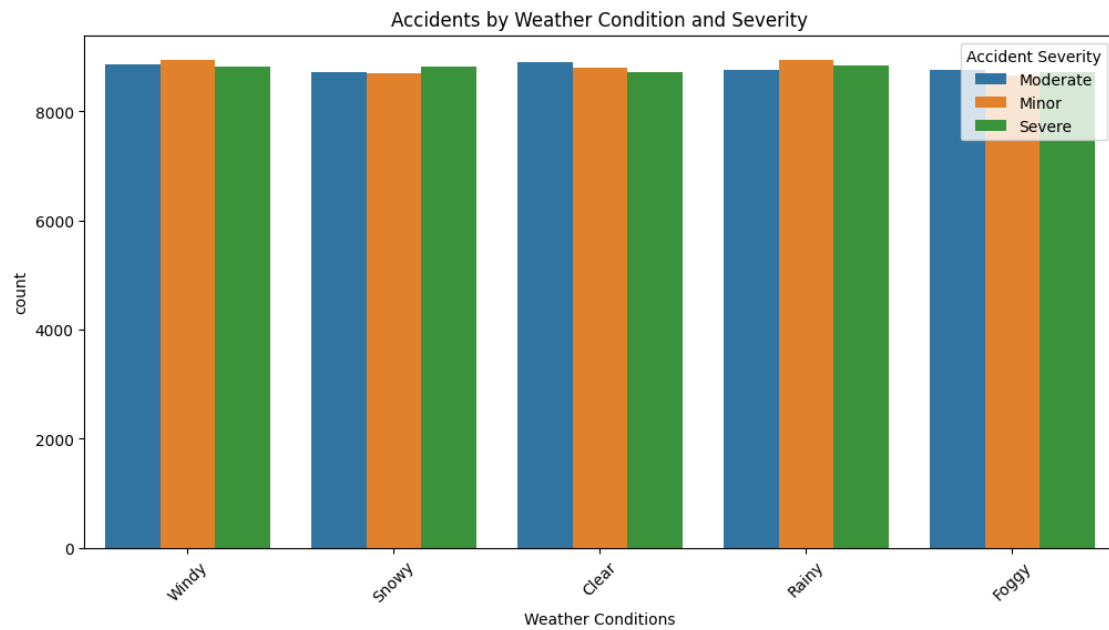
0.0.4 Question 4

- a. Plot a chart showing accidents by Weather Condition, Road Type and Accident Severity

```
[51]: # Grouped bar plot
plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='Weather Conditions', hue='Accident Severity')
plt.title("Accidents by Weather Condition and Severity")
plt.xticks(rotation=45)
plt.show()

# Stacked bar plot for Road Type and Severity
plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='Road Type', hue='Accident Severity')
```

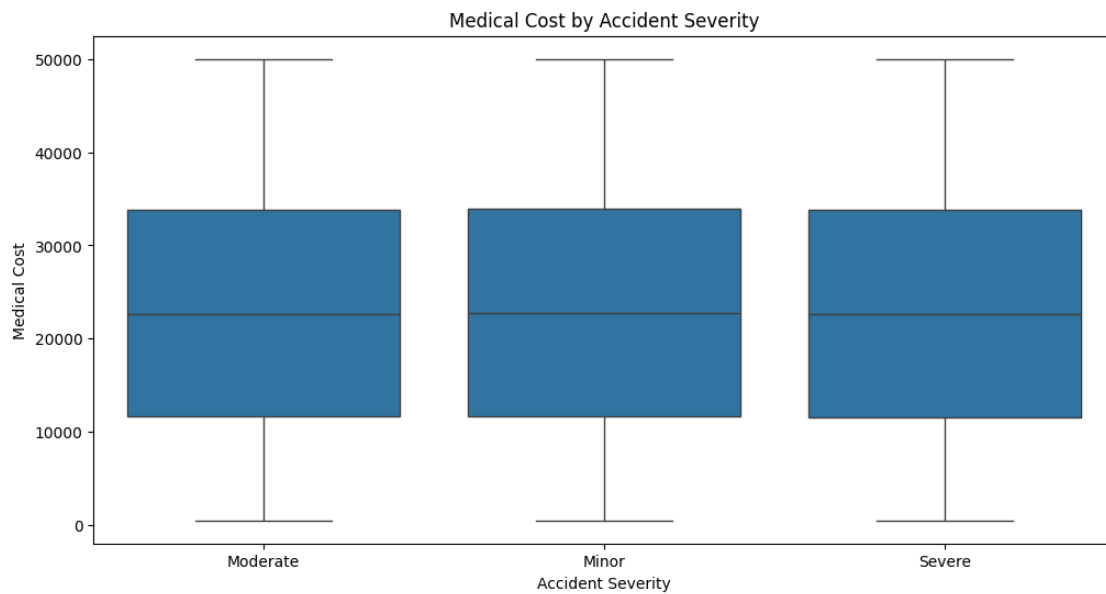
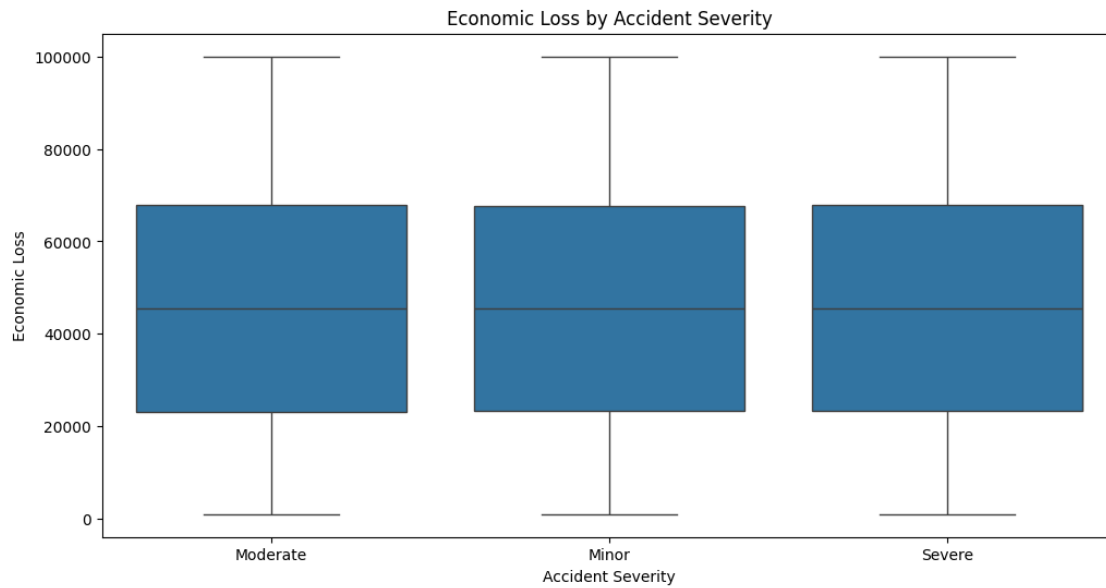
```
plt.title("Accidents by Road Type and Severity")
plt.xticks(rotation=45)
plt.show()
```



b. Plot a boxplot for Economic Loss, Medical Cost grouped by Accident Severity

```
[33]: plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x='Accident Severity', y='Economic Loss')
plt.title("Economic Loss by Accident Severity")
plt.show()

plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x='Accident Severity', y='Medical Cost')
plt.title("Medical Cost by Accident Severity")
plt.show()
```



c. Plot a scatter plot between Traffic Volume and Accident Severity / Economic Loss

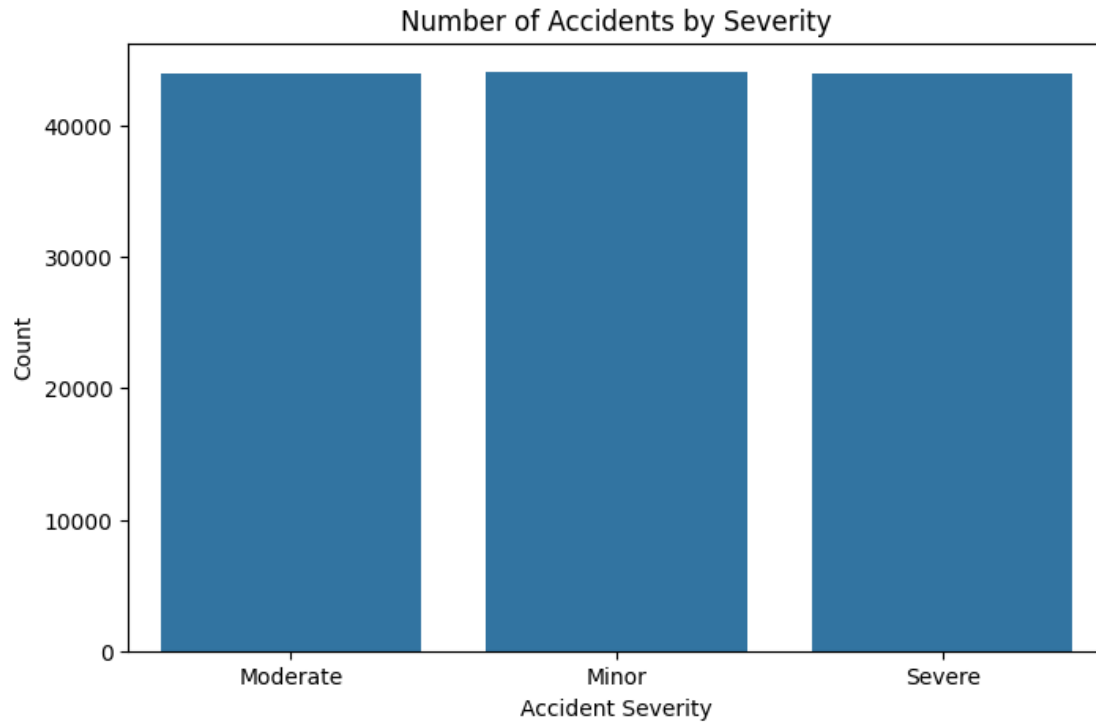
```
[ ]: plt.figure(figsize=(12, 6))
sns.scatterplot(data=df, x='Traffic Volume', y='Economic Loss', hue='Accident_Severity')
plt.title("Traffic Volume vs. Economic Loss (Colored by Severity)")
plt.show()
```



d. Interpret any 3 charts and explain what they reveal about accident patterns

```
[84]: # Bar Chart: Number of Accidents by Accident_Severity

plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Accident Severity')
plt.title("Number of Accidents by Severity")
plt.xlabel("Accident Severity")
plt.ylabel("Count")
plt.show()
```

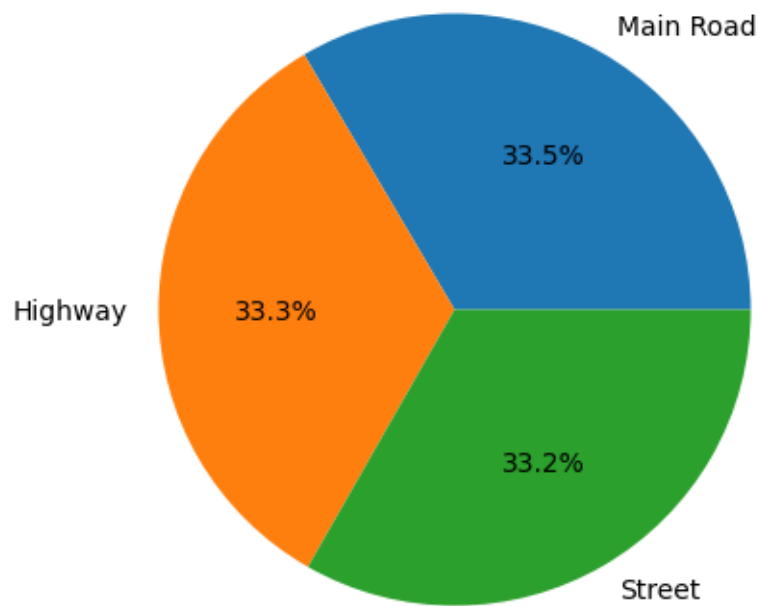


Accident Pattern: Pattern: This chart shows the frequency of accidents for each severity level - Moderate, Minor, Sever

```
[45]: # Pie Chart: Distribution of Accidents by Road_Type

plt.figure(figsize=(8, 5))
df['Road Type'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Distribution of Accidents by Road Type")
plt.ylabel("")
plt.show()
```

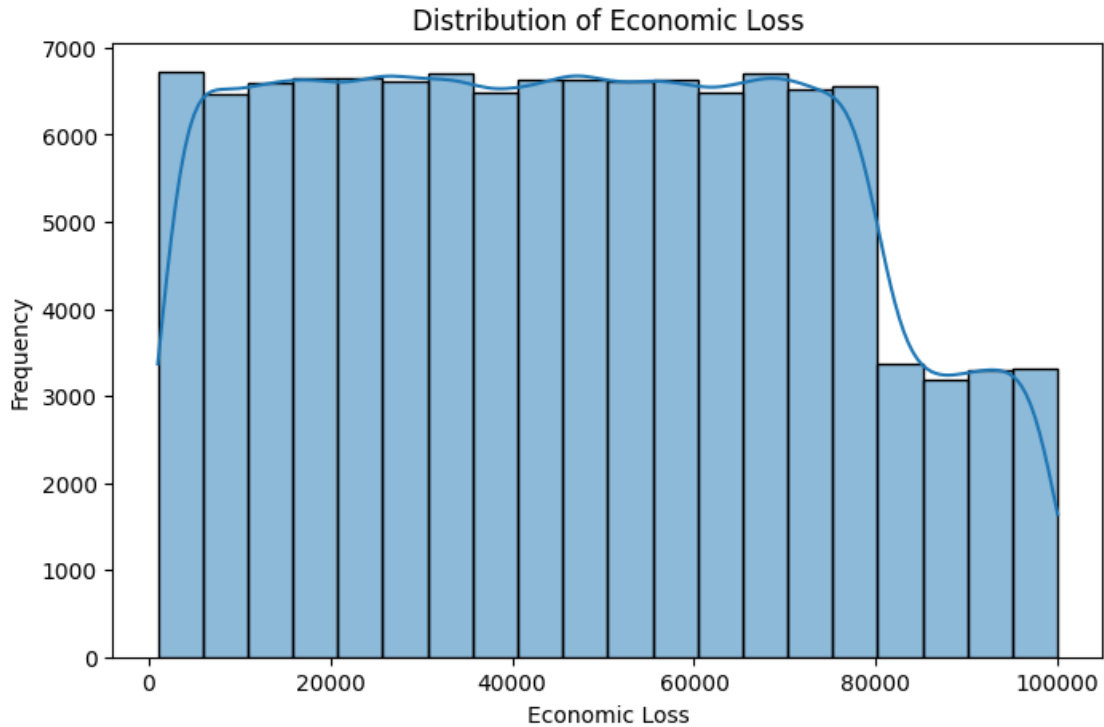
Distribution of Accidents by Road Type



Accident Pattern: This chart shows the proportion of accidents across different road types - Main Road, Highway, Street

```
[49]: # Histogram: Distribution of Economic_Loss

plt.figure(figsize=(8, 5))
sns.histplot(data=df, x='Economic Loss', kde=True, bins=20)
plt.title("Distribution of Economic Loss")
plt.xlabel("Economic Loss")
plt.ylabel("Frequency")
plt.show()
```



Accident Pattern: This chart shows how economic loss is distributed across accidents

0.0.5 Question 5

Perform a correlation analysis on the numerical variables by creating a correlation matrix.

Identify the strength and direction of the correlations, and interpret the relationships between Emergency Response Time, Medical Cost, Economic Loss, Driver Alcohol Level, and Accident Severity.

Further, explain why categorical variables should not be directly included in correlation analysis without appropriate encoding.

```
[62]: # Convert Accident Severity to Severity_Num

df['Severity_Num'] = df['Accident Severity'].map({'Minor': 1, 'Moderate': 2, 'Severe': 3})
```

```
[63]: # Select key numerical columns

corr_cols = ['Emergency Response Time', 'Medical Cost', 'Economic Loss',
             'Driver Alcohol Level', 'Severity_Num', 'Traffic Volume']

corr_matrix = df[corr_cols].corr()
print(corr_matrix)
```

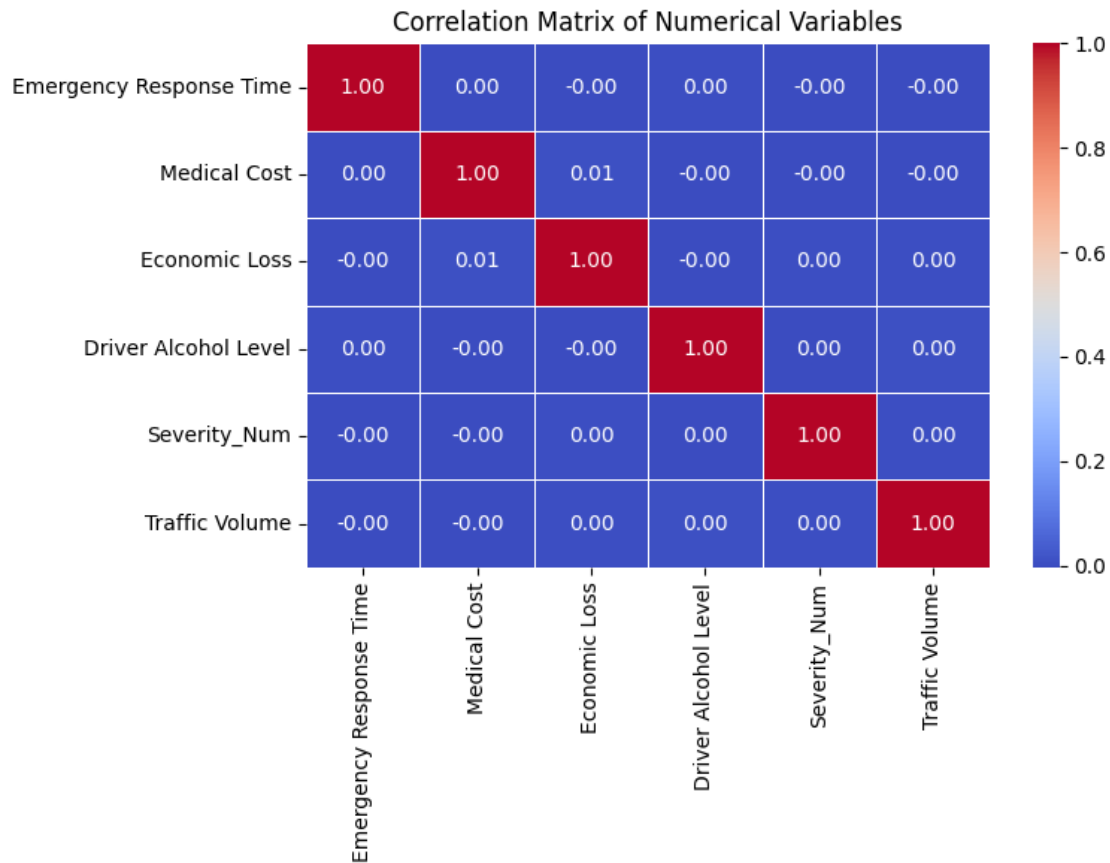
```

# Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Matrix of Numerical Variables')
plt.tight_layout()
plt.show()

```

	Emergency Response Time	Medical Cost	Economic Loss	\
Emergency Response Time	1.000000	0.000265	-0.000455	
Medical Cost	0.000265	1.000000	0.005073	
Economic Loss	-0.000455	0.005073	1.000000	
Driver Alcohol Level	0.000784	-0.001417	-0.002992	
Severity_Num	-0.000578	-0.002917	0.000517	
Traffic Volume	-0.001719	-0.003406	0.002142	

	Driver Alcohol Level	Severity_Num	Traffic Volume
Emergency Response Time	0.000784	-0.000578	-0.001719
Medical Cost	-0.001417	-0.002917	-0.003406
Economic Loss	-0.002992	0.000517	0.002142
Driver Alcohol Level	1.000000	0.000052	0.004581
Severity_Num	0.000052	1.000000	0.001350
Traffic Volume	0.004581	0.001350	1.000000



Interpretation:

- Medical Cost & Economic Loss: likely positive correlation (more cost = more loss)
- Driver Alcohol Level & Severity: positive (higher alcohol → more severe accidents)
- Emergency Response Time & Severity: may be positive (severe accidents take longer)
- Categorical variables (Weather, Road Type etc.) cannot be directly correlated without encoding (Label/One-Hot) as correlation requires numeric values.

0.0.6 Question 6

- Use the IQR method to identify outliers for Medical Cost, Economic Loss, Traffic Volume and Emergency Response Time

```
[64]: # a. IQR Outlier Detection
def detect_outliers_iqr(col):
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
```

```

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df[col] < lower) | (df[col] > upper)]
print(f"{col}: Q1={Q1:.2f}, Q3={Q3:.2f}, IQR={IQR:.2f}, "
      f"Lower={lower:.2f}, Upper={upper:.2f}, Outliers={len(outliers)}")
return outliers

for col in ['Medical Cost', 'Economic Loss', 'Traffic Volume', 'Emergency_
↳Response Time']:
    detect_outliers_iqr(col)

```

Medical Cost: Q1=11594.70, Q3=33869.68, IQR=22274.98, Lower=-21817.77, Upper=67282.14, Outliers=0
Economic Loss: Q1=23253.53, Q3=67731.98, IQR=44478.45, Lower=-43464.15, Upper=134449.66, Outliers=0
Traffic Volume: Q1=2554.84, Q3=7523.93, IQR=4969.09, Lower=-4898.81, Upper=14977.57, Outliers=0
Emergency Response Time: Q1=18.69, Q3=46.29, IQR=27.59, Lower=-22.70, Upper=87.68, Outliers=0

b. Examine whether high Medical Cost values are errors or meaningful extreme cases

High Medical Cost Interpretation: - Road accidents can genuinely result in very high medical bills (ICU, surgeries, long recovery) - These are likely MEANINGFUL EXTREME CASES, not data errors - Cross-check: if Accident Severity = 'Severe' and Fatalities > 0 → high cost is justified - Should NOT be removed; consider robust models or log-transformation instead

```

[65]: # Verify: check medical cost for severe accidents
print(df.groupby('Accident Severity')['Medical Cost'].describe())

```

	count	mean	std	min \
Accident Severity				
Minor	39680.0	23045.180923	13319.387433	500.110091
Moderate	39644.0	22989.816355	13289.106973	501.920576
Severe	27655.0	22947.166688	13326.086840	501.365848

	25%	50%	75%	max
Accident Severity				
Minor	11627.283537	22797.664165	33959.671985	49996.85700
Moderate	11612.436848	22680.063880	33808.292763	49998.99995
Severe	11532.162270	22583.939920	33834.995050	49999.93013

0.0.7 Question 7

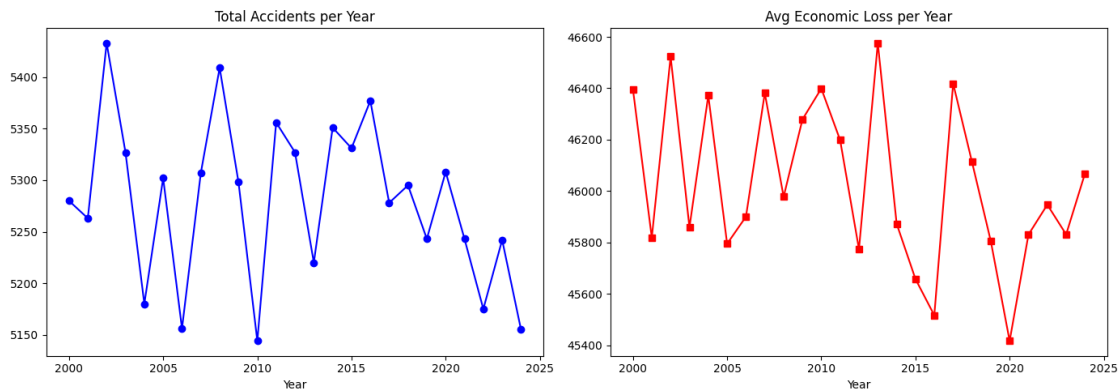
a. How do accident trends and impact metrics vary over time?

Ans.

```
[66]: yearly = df.groupby('Year').agg(
    Total_Accidents=('Accident Severity', 'count'),
    Avg_Medical_Cost=('Medical Cost', 'mean'),
    Avg_Economic_Loss=('Economic Loss', 'mean'),
    Avg_Fatalities=('Number of Fatalities', 'mean')
).reset_index()

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(yearly['Year'], yearly['Total_Accidents'], marker='o',
    color='blue')
axes[0].set_title('Total Accidents per Year')
axes[0].set_xlabel('Year')

axes[1].plot(yearly['Year'], yearly['Avg_Economic_Loss'], marker='s',
    color='red')
axes[1].set_title('Avg Economic Loss per Year')
axes[1].set_xlabel('Year')
plt.tight_layout()
plt.show()
```



b. How do environmental and road conditions contribute to accident outcomes?

```
[70]: print("\nAvg Fatalities by Weather Condition:")
print(df.groupby('Weather Conditions')['Number of Fatalities'].mean().
    sort_values(ascending=False))

print("\nAvg Severity encoding by Road Type:")
print(df.groupby('Road Type')['Severity_Num'].mean().
    sort_values(ascending=False))
```

```
Avg Fatalities by Weather Condition:
Weather Conditions
Foggy      2.005165
```



```

Rainy      2.003087
Clear      1.995648
Windy      1.987531
Snowy      1.985828
Name: Number of Fatalities, dtype: float64

```

```

Avg Severity encoding by Road Type:
Road Type
Street      1.999180
Main Road   1.999072
Highway     1.998839
Name: Severity_Num, dtype: float64

```

c. What factors most significantly influence accident severity and overall impact?

```

[69]: print("\nCorrelation with Severity:")
      factors = ['Driver Alcohol Level', 'Speed Limit', 'Traffic Volume',
                  'Emergency Response Time', 'Number of Vehicles Involved']
      print(df[factors + ['Severity_Num']].corr()['Severity_Num'].
            ↪sort_values(ascending=False))

```

```

Correlation with Severity:
Severity_Num      1.000000
Number of Vehicles Involved  0.003024
Traffic Volume      0.001350
Speed Limit        0.001313
Driver Alcohol Level  0.000052
Emergency Response Time -0.000578
Name: Severity_Num, dtype: float64

```

d. How do accident characteristics and impacts vary geographically?

```

[80]: print("\nAccidents and Economic Loss by Region:")
      print(df.groupby('Region').agg(
          Count=('Accident Severity', 'count'),
          Avg_Economic_Loss=('Economic Loss', 'mean'),
          Avg_Fatalities=('Number of Fatalities', 'mean')
      ).sort_values('Avg_Economic_Loss', ascending=False))

      # Bar chart for regional comparison
      region_data = df.groupby('Region')['Economic Loss'].mean()
      region_data.plot(kind='bar', color='teal', figsize=(8, 20))
      plt.title('Avg Economic Loss by Region')
      plt.ylabel('Economic Loss')
      plt.tight_layout()
      plt.show()

```

Accidents and Economic Loss by Region:

	Count	Avg_Economic_Loss	Avg_Fatalities
Region			
Asia	26351	46344.283107	1.990437
South America	26264	46125.529702	1.990900
Europe	26345	46010.872320	2.005846
North America	26415	45874.133514	1.997729
Australia	26625	45789.300014	1.992300

